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Research Grant Application Form – Short Proposal Performa

Submission deadline: 30/10/2026 at 11:59 PM. Saved applications will not be considered after this time.

1. General Information

Principal Investigator	
CNIC	
University	
Title of the Research *	A Hierarchical Multi Branch Attention Network for Interpretable Multi Stage Cyber attack Detection in the Internet of Medical Things
Industrial Collaborator (Name & Organisation) *	Hospital IT/security Branch, Medical Device manufacturer or cyber security firm **Concrete contribution:** (i) Controlled access to anonymized or representative IoMT network traffic characteristics and device context for validation (ii) A testbed or sandbox environment to check the model on traffic beyond MedSec25 (iii) Feedback from security analysts on the usefulness of phase labels and SHAP explanations (iv) Domain expertise on clinical device threat scenarios.
International Collaborator *	
Total Duration (in Months) *	24 months (development, leakage resistant evaluation, explainability analysis and collaborator validation)
Theme of Proposed Research / Research Domain *	Cyber Security / Artificial Intelligence for Critical Infrastructure - Digital Health Security
Discipline of Proposed Research *	Computer Science
Sub Discipline *	Information Security (Artificial Intelligence – Cyber Security)

2. Proposal Details

Sub-Field along with Suggested Reviewer Domain *

Sub-Field: Cyber security, specifically machine learning-based intrusion detection for the Internet of Medical Things (IoMT), with a focus on multi-view deep learning and explainable AI (XAI) for healthcare network security.

Suggested Reviewer Domain

Reviewers with expertise in (i) IoMT/IoT intrusion detection and healthcare network security, (ii) hierarchical or multi-branch neural architectures for network-traffic analysis, and (iii) explainable AI for security applications (e.g., SHAP-based analysis).

Problem Statement *

Maximum 500 words

Hospitals now run on connected devices like infusion pumps, ventilators, cardiac monitors and EEG headsets, all sharing the same network. That convenience comes at a cost. A successful attack on these systems can put patients at risk, not just expose data. The devices often run embedded software that can not be changed, rely on older protocols like HL7 and DICOM, and sit on network segments that let an attacker move sideways with little resistance. Standard enterprise intrusion detection tools were not built for this environment.

Current research on IoMT intrusion detection falls short in three connected ways.

First, most systems only answer “is this traffic an attack or not?” They cannot tell a quiet reconnaissance scan from the final stage of data theft, so defenders miss the chance to step in early.

Second, most models look at the data from one angle only. Flow statistics and the order in which flows arrive each carry useful information, but they are rarely combined. Tree-based models are also seldom merged with neural networks in a way that lets the system learn which one to trust.

Third, explainability is usually bolted on at the end, if its included at all. Hospitals and regulators need alerts that analysts can understand and audit, and a black box model is hard to defend in that setting.

Many studies also evaluate on general-purpose IoT or enterprise datasets that do not resemble a hospital. MedSec-25 is a better fit: it was collected on a physical IoMT testbed (ECG, EEG, respiration and environmental sensors on Raspberry Pi nodes) and labels a four-phase, MITRE ATT&CK-aligned campaign (Reconnaissance, Initial Access, Lateral Movement, Exfiltration) next to benign traffic. These phase labels make phase-level classification possible, yet to the best of our knowledge no classifier has used them to combine hierarchical phase detection with multi-view fusion. Evaluation practice in this area also does not always guard against leakage from oversampling, scaling or feature selection applied before the data split.

The problem therefore remains open: no existing system jointly delivers phase-level classification, multi-branch fusion of tabular, temporal and tree-based views, and clinician-usable explanations on MedSec-25.

Research question: Can a hierarchical, phase-aware multi-branch network with cross-attention fusion and dual-level SHAP explanations classify IoMT attack phases on MedSec-25 more accurately than single-view baselines under a leakage-resistant evaluation protocol, while giving clinical security teams explanations they can act on?

Objectives and Scope *

Maximum 500 words

Objectives

O1. The project will design and implement HMBAN, a hierarchical architecture with three parallel branches: a TabTransformer for feature interactions, a BiLSTM with temporal attention for flow sequence behaviour, and an XGBoost leaf embedding branch for sharp decision boundaries (target: working prototype by month 8).

O2. The project will develop a learnable cross-attention layer that fuses the three branches and records per prediction branch weights, and will use ablations (removing each branch and the fusion layer) to test whether fusion beats every single branch on macro F1.

O3. The project will implement a two stage classifier: a high recall benign-versus-malicious gate (initial threshold 0.35, to be tuned on validation folds against a recall target fixed in advance), followed by a four class phase classifier for Reconnaissance, Initial Access, Lateral Movement and Exfiltration.

O4. The project will add SHAP explanations at global level (feature importance per phase) and local level (why a given alert fired), and will assess their usefulness through at least one structured feedback session with the industrial collaborators security staff.

O5. The project will compare HMBAN against four baselines (standalone TabTransformer, BiLSTM, XGBoost and a CNN-based IDS) using accuracy, macro precision, recall and F1, ROC-AUC, false positive rate and false negative rate, reported as mean standard deviation over five folds with a paired significance test.

O6. The project will measure inference time on GPU and CPU, and the cost of generating SHAP explanations, to judge suitability for batch flow analysis and post hoc audit.

Scope

Included – data: MedSec25 only (about 554,000 labelled flows, 84 features, reduced to about 40 selected features).

Included – evaluation: a controlled, leakage resistant protocol: 80/20 stratified split with 5 fold cross validation, where SMOTE, scaling, feature selection and XGBoost leaf embeddings are fitted inside each training fold only.

Included – level of classification: per flow attack phase classification aligned with MITRE ATT&CK. Session level tracking of how an attack progresses over time is excluded.

Explicitly left for future work: edge device deployment, federated training across hospitals, adversarial robustness testing, session-level tracking, and phases absent from MedSec25 (Execution, Persistence, Impact)

Research Significance and the Challenges Addressed *

Maximum 500 words

Hospitals depend so heavily on connected medical devices that a network attack can endanger patients, not just their data. The main contribution of this project is methodological hierarchical phase aware detection combined with multi view fusion and builtin explainability tailored to clinical security teams. Instead of stopping at attack or not, the proposed HMBAN will sort malicious traffic into four stages (Reconnaissance, Initial Access, Lateral Movement and Exfiltration), so that a response team can act on what is actually happening.

Phase awareness supports early intervention in practice, a team could isolate a scanning host at the reconnaissance stage, before any credentials are stolen or check recent logins across a subnet once

lateral movement appears. Each alert will carry SHAP explanations and attention weights so analysts can see why it was raised. This transparency responds to regulatory expectations for explainable AI, such as the EU AI Act and medical device software standards.

The project will address several challenges explicitly. Class imbalance (Reconnaissance alone is about 72% of MedSec25) will be handled with fold wise SMOTE and macro averaged metrics. Data leakage will be controlled by fitting every pre processing step inside training folds. Alert fatigue will be tackled by combining a high recall gate with phase labels and explanations that help analysts prioritize. Confusion between similar late stage attacks will be tackled by the second stage and multi branch fusion. Limited real hospital validation is a further challenge, results from one testbed may not transfer, so the industrial collaborator will review outputs and explanations on representative traffic where access is agreed.

The limitations will be reported openly a single testbed, per flow rather than session level analysis, CPU inference that may be too slow for packet level use, and SHAP being better suited to post hoc audits than to explaining every alert in real time. Performance figures appear only as targets under expected outcomes.

Research Methodology *

Maximum 1000 words

The project will follow a planned four stage experimental pipeline, pre processing, three parallel feature extraction branches, cross attention fusion and a two phase hierarchical classifier with SHAP explanations.

Datasets and pre processing. The study will use MedSec25, collected on a physical IoMT testbed of Raspberry Pi nodes with ECG, EEG, respiration and environmental sensors. It contains about 554,000 labelled flows described by 84 CICFlowMeter features, covering benign traffic and a four phase MITRE ATT&CK aligned campaign. Identifier columns (Flow ID, IP addresses, timestamp) will be excluded from the model inputs and duplicate records removed, source IP and timestamp will be used only to order and group flows for the temporal branch. An 80/20 stratified split will be made, with 5 fold cross validation on the training portion. Minimum maximum scaling, SMOTE and feature selection will be fitted only on each folds training data, so that no information reaches validation or test data. After near constant features are removed, mutual information and Recursive Feature Elimination will reduce the set to about 40 features (inter arrival times, packet sizes, flag counts, flow rates).

Branch 1: TabTransformer. The Protocol field will be embedded and each of the 39 numerical features projected into its own token, giving a 40 token sequence. Multi head self attention will learn which feature combinations separate the attack phases and will output a 256 dimensional vector.

Branch 2: BiLSTM. Flows will be sorted by time, grouped by source IP and cut into windows of 32 flows, with zero padding and an attention mask for shorter windows. A two layer BiLSTM will read each window, and a temporal attention layer will compress it into a 256 dimensional summary.

Branch 3: XGBoost. A 200 tree ensemble will be trained with a regularized cross entropy objective. Each sample leaf index embedding will be projected to 256 dimensions; training embeddings will come from out of fold predictions so that this branch cannot encode its own training labels.

Cross-attention fusion. A learnable query vector will attend over the three branch outputs, producing weights that sum to one and a fused 256dimensional representation. The weights will be logged at inference to show which branch drove each prediction.

Hierarchical classification. Phase 1 will be a two layer MLP separating benign from malicious traffic, with an initial threshold of 0.35 (tuned on validation folds) so that it acts as a high recall gate. Phase 2 will be a four class softmax classifier trained only on malicious samples, so it can concentrate on differences between attack phases.

Explainability. SHAP values will be computed on a random subset of about 5000 test instances, using bees warm plots for global importance and waterfall plots for individual predictions. This runs after inference and will not slow detection.

Computing resources. The project will use an NVIDIA A100 GPU with PyTorch 2.2.0, available to the research team for training CPU only inference will be timed on a standard workstation.

Risks and mitigation. Single testbed results may not generalize, so conclusions will be limited to MedSec25 and checked with the industrial collaborator on representative traffic where access is agreed. Computational requirements training three branches plus SHAP is costly, so SHAP will run on a sampled subset and hyperparameter search will be kept bounded. Class imbalance fold-wise SMOTE, macro averaged metrics and per class reporting will be used and the effect of oversampling will be checked against a class weighted alternative.

Expected Outcomes *

Maximum 500 words

The project is expected to deliver a practical intrusion detection approach for IoMT networks that goes beyond attack or not classification. The primary output will be HMBAN, which combines a TabTransformer, a BiLSTM and an XGBoost branch through cross attention and classifies traffic in two stages benign versus malicious first, then attack phase (Reconnaissance, Initial Access, Lateral Movement or Exfiltration)

Performance targets (anticipated, not yet achieved) HMBAN is expected to outperform each individual branch and the CNN-IDS baseline on MedSec25, with targets of a macro F1 of about 99%, a ROC-AUC of about 0.999, and false positive and false negative rates below roughly 1%. The second stage is expected to separate similar phases (Lateral Movement and Exfiltration) better than single model solutions. These targets will be revised to whatever the cross validated results show.

Deliverables. (1) The HMBAN model and code

(2) A leakage resistant benchmark on MedSec25 in which every model shares the same pre processing and all fitting happens inside training folds

(3) Global and local SHAP analyses identifying the features behind each decision (Flow IAT Mean, SYN Flag Count)

(4) Inference time measurements with a target of a few milliseconds per sample on GPU and under about 20 ms on CPU, suitable for batch flow analysis

(5) A limitations and future work discussion.

Academic outputs. One to two peer reviewed papers (architecture and benchmark, explainability and collaborator feedback) one technical report and training of at least one MS student in IoMT security and explainable AI.

Code and model release. Code and trained models will be released in a public repository after publication, subject to MedSec25 licensing terms and any agreement with the industrial collaborator. Where data cannot be redistributed, scripts and configuration files will be shared instead.

Limitations will be reported honestly: a single testbed, heavy class imbalance, per flow rather than session level classification and CPU inference speed that may not suit packet level use. These point

to future work on edge deployment, federated training, adversarial testing and broader attack phase coverage.

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